Benchmarking LLM Integration Strategies for MCDA-Assisted Household Energy Decision Support

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Abstract

Households contribute 13% of U.S. carbon emissions, yet most people rely on biased decision making rather than tailored choices. Multi-Attribute Value Theory (MAVT) offers a rigorous method of Multi-Criteria Decision Analysis (MCDA) to reduce emissions, but building MCDAs daily requires expertise that limits household adoption. This study evaluated whether LLMs can bridge that gap by comparing three architectures: Pure Prompting, Retrieval-Augmented Generation (RAG-Enhanced) Scoring, and a Hybrid of LLM parameter extraction with calculation. Ground truth calculators were built to benchmark MAVT scores across HVAC, appliance, and shower decisions. *[Quantitative claims in this paragraph are BLOCKED pending corrected multi-model results; see audit log.]*

Keywords: Large language models, Multi-criteria decision analysis, Multi-Attribute Value Theory, Household energy, Decision support systems, Retrieval-Augmented Generation, Sustainability

1. Introduction

The average U.S. household emits roughly 8,744 pounds of CO_2e and consumes about 10,791 kWh of electricity each year [1, 2]. However, there is more than enough evidence that this could be much further optimized to reduce cost to families and environmental impact [ADD SOURCES]

Tools that combine optimization, behavioral insight, and decision support can close part of that gap, but only if they are usable without specialized training. Certain frameworks, such as Multi-Criteria Decision Analysis (MCDA), can be used to choose an optimal balance of factors that matter to users [10]. Unfortunately, these are extremely taxing to create oneself accurately even as an expert, and they require extensive time which makes using them by themselves impractical for household decisions.

However, it has been proposed in that past that LLMs could be used to automate this process, thus making it practical for household decisions. However, without knowing how accurate these LLMs are as experts, it is unclear whether this would be a beneficial or detrimental idea.

Modern LLMs carry general knowledge about the energy use and carbon intensity of common products and behaviours, and many systems can retrieve additional information at inference time. They have not, however, been rigorously evaluated for individual-level household decisions against a deterministic, physics-based ground truth, nor has it been established which way of integrating an LLM into the MCDA workflow—direct prompting, retrieval-augmented prompting, or LLM extraction feeding a deterministic calculator—best reproduces that ground truth. This study addresses that gap by benchmarking the three integration strategies head-to-head across three household energy decision types, holding the model and the MAVT weighting constant so that the integration strategy is the variable under test.

This study addresses the following research question: which LLM integration architecture—Pure Prompting, RAG-Enhanced, or Hybrid—most accurately replicates physics-based MAVT scoring for household energy decisions, and how well this advantage hold across a models of varying intelligence.

2. Literature Review

When transportation, food, and goods are folded in, per-capita footprints rise to about 17.9 tons of CO_2e and 81,767 kWh of total annual energy [3, 4]. The U.S. Department of Energy estimates that households can save \$200–\$400 annually by adopting ENERGY STAR practices [5, 6]. A persistent gap between intent and behavior keeps many households from realizing those savings and footprint reduction[9].

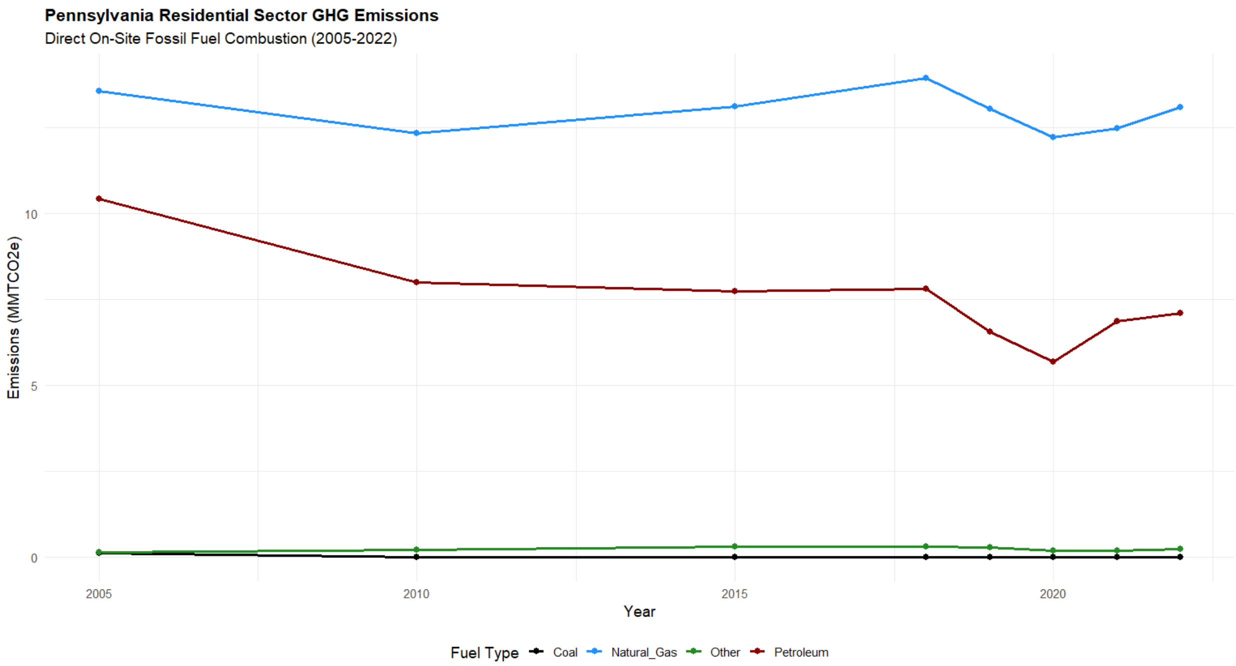

Figure 1: Residential emissions over the past 50 years (EIA, 2022) [7, 8].

Attari et al. (2010) found that consumers undervalue energy use and savings by a factor of 2.8 on average, systematically underestimating high-energy activities while overestimating low-energy ones [26]. Energy labels partially address this, but effectiveness depends on framing: Newell and Siikamäki (2014) found willingness to pay for efficient products rose when savings were clearly displayed, while Allcott and Taubinsky (2015) reported a 12% market-share increase for compact fluorescent bulbs when benefits were made visible—though gains collapsed once cost projections became ambiguous [48, 49]. Label design also matters: the EU label underperformed relative to U.S. equivalents owing to poor clarity and limited consumer trust [50].

hile the average U.S. household consumes 10,791 kWh annually, this consumption is not evenly distributed. Space heating accounts for approximately 43% of residential energy use, followed by appliances (13%), water heating (12%), and air conditioning (8%) [**?**]. This distribution highlights key intervention points for emissions reduction. National reform of certain key household practices could result in a 123 million metric tonne (MtC) reduction in carbon year as of 2005, when residential emissions accounted for 38% of overall US CO2 emissions, or 626 MtC. Additionally, these changes would result in little to no reduction in satisfaction, a necessary factor in widespread implementation [**?**].

Figure 2 reproduces the Dietz et al. display of behavioral plasticity, net carbon effect, and the Reasonably Achievable Emission Reduction (RAER) per action.

Dietz et al. [**?**] also demonstrated that household-level behavioral changes, if widely adopted, could cut U.S. emissions by 123 MtC annually. However, concerns for behavioral plasticity—how likely people are to implement each action based on financial and social incentives—remain an important factor [**?**]. Actions such as installing programmable thermostats demonstrate high emissions reductions and adoption likelihood, whereas vehicle upgrades show reductions but low likelihood. More often than not, individuals pointed to curtailment rather than implementing

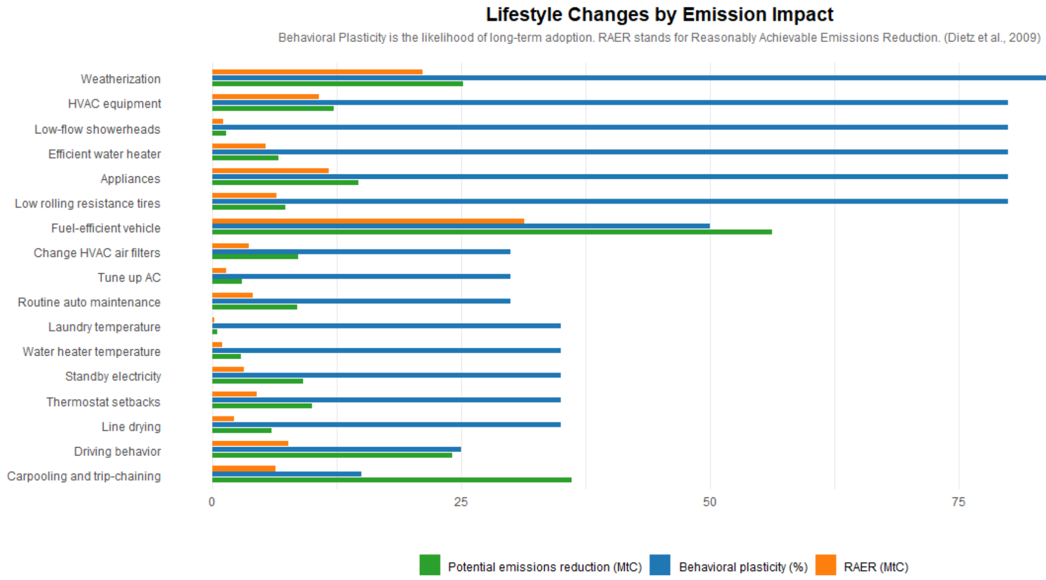

Figure 2: Household behavior changes and their emissions impact (after Dietz et al., 2009 [14]).

new products for energy efficiency in their households [?]. Thus, targeting simple, heuristic actions is likely to be more effective at this scale than recommending car or HVAC replacements, even if those may be more impactful.

A practical obstacle for MCDA-style tools at the household level is the workflow tax they impose on users. Even when the benefits are clear, switching from a routine choice to a model-mediated decision adds time and cognitive cost, both of which are documented deterrents to adoption [15, 16, 17]. Most household decisions in this space, including thermostat setpoints, appliance run times, and shower length, are heuristic rather than analytical.

Heuristic decision-making is efficient for everyday choices but degrades in complex or high-stakes settings. Under time pressure, people default to satisficing, choosing the first option that clears a minimum threshold, or to dominant prior behaviors, even when better alternatives exist [18]. When time pressure combines with information overload, judgment error rates rise meaningfully [19, 20]. Household energy decisions typically occur in exactly this regime: short deliberation windows, competing demands, and incomplete information. Tools that require sustained attention or structured inputs are unlikely to influence behavior under those conditions.

An interesting phenomenon is when consumers were asked to guess the energy savings of products, they undervalued "energy use and savings by a factor of 2.8 on average, with small overestimates for low-energy activities and large underestimates for high-energy activities," so energy labels are indeed helpful [?]. A study by Allcott, Hunt, and Dmitry Taubinsky found that willingness to pay increased by $2.30 for compact fluorescent lights in particular, and market share increased by 12% when the benefits were displayed. They proposed that this would extend to most other household appliances. It is important to note that once the statistics on cost savings were unclear, the impact of all information dropped significantly [?]. Thus, any tool that was used to recommend decisions should surely take into account energy cost and show some sort of projected change to help motivate users, and make it more likely that they continue following recommendations.

However, a nuance to that is the compounding obstacle of temporal discounting: the perceived value of a future reward decreases with delay in a non-uniform way. People prefer $100 today over $110 tomorrow, yet prefer $110 in a year to $100 in eleven months, suggesting the discount operates sharply over short intervals [?]. The benefits of reducing energy consumption are largely abstract and deferred; even tools that promise genuine future savings may be ignored if those benefits are not immediate, concrete, or personally meaningful.

Multi-criteria decision analysis (MCDA), life-cycle assessment (LCA), and Pareto-based prediction methods have been used to track and reduce environmental impact at building and organizational scales [10, 11]. Engineering frameworks of this kind can optimize complex decisions, but their construction is time-consuming: signal-processing pipelines such as the Hilbert–Huang transform are not rapid, and constructing the decision matrix itself is a recognized bottleneck [12]. Huang and Burgherr (2024) addressed part of this by building a streamlined MCDA calculator that compresses several accepted methods into a single workflow [13]. Even with their simplification, the user still needs enough familiarity with MCDA construction to frame the alternatives, criteria, and weights, which keeps the tool out of routine household use.

The harder problem is operational. Constructing an MCDA framework for a specific decision normally requires methodological knowledge, including the choice of weighting scheme, value function form, and normalization, alongside rubric design and domain familiarity [12, 13]. Existing simplified calculators reduce but do not eliminate this barrier. A large language model that can convert a natural-language description of a household situation into the alternatives, criteria, and contextual parameters that an MCDA needs would, in principle, remove the methodological barrier altogether.

AI is increasingly being used to build or operate decision frameworks in domains that share the household setting's combination of variability and judgment. Generative AI now appears in operations and supply-chain decision pipelines [21], in marketing strategy [22], and in clinical decision support [23]. These techniques have been used across many different industries, and AI's ability to manage large datasets suggests it could scale down to personal decision-making as well.

Specific AI techniques integrated into MCDA pipelines include fuzzy systems, [ADD NEW CITATIONS THAT WE HAVE ON FUZZY] which account for uncertainty and degrees of truth to approximate human reasoning. This has the potential to be an effective way to help guide a model that is given no outside knowledge, such as Pure-Prompting. [**?** **?**].

The plausibility of this approach is reinforced by recent work outside the energy domain. Svoboda and Lande (2024) used GPT-4 to construct an analytic hierarchy process (AHP) decision model for cybersecurity, reporting a consistency ratio below the conventional 0.10 acceptability threshold and noting that earlier models such as GPT-3 were unreliable [24]. That result indicates that LLMs above a certain capability level, can produce coherent MCDA structures, but it leaves open the separate question of whether they can produce accurate quantitative scores against a physics-based ground truth.

LMs are autoregressive next-token predictors, not symbolic computation engines. Cobbe et al. (2021) introduced the GSM8K benchmark of grade-school arithmetic word problems and found that even large Transformers fall well short of human accuracy once a problem needs more than a few chained steps, with errors compounding rather than cancelling [**?**]. The gap hasn't closed in newer models: Li et al. (2025) test GPT-4, DeepSeek, and others on basic arithmetic, magnitude comparison, and multi-step numeric reasoning, and report the same weaknesses despite strong general-language performance [**?**]. Imani et al. (2023) approach the same problem from the fix side, building MathPrompter on the observation that LLMs "often provide incorrect answers" on arithmetic benchmarks unless cross-checked against multiple generated solution paths [**?**]. Approximating a tent-shaped comfort function, applying a multiplicative budget penalty, and tracking a TOU emissions schedule, then reporting a single calibrated number, is exactly this kind of task – so unaided LLM scoring (the Pure Prompting architecture here) carries a structural rather than incidental risk of numeric drift. The canonical use of Retrieval-Augmented Generation is injecting external facts into an LLM's context to reduce hallucination in open-domain QA [25]. An MCDA scoring task is closer to Case-Based Reasoning, which treats problem-solving as retrieving similar prior cases and reusing their recorded solutions [**?**]. Wiratunga et al. (2024) build CBR-RAG, where retrieved legal cases act as structured exemplars that shape an LLM's reasoning rather than supply facts [**?**]; Minor and Kaucher (2024) get a similar result, finding that retrieved example explanations used as in-context demonstrations improve LLM-generated outputs over zero-shot prompting [**?**]. Read against this, the RAG-Enhanced architecture's exemplars look less like supplementary facts and more like a case-based rubric: scenarios whose ground-truth scores anchor the model's numeric judgment to the decision space's actual reference distribution, rather than whatever scale it absorbed during pretraining. There's a third option, and it sidesteps the arithmetic weakness instead of mitigating it: keep the LLM on language understanding, hand the math to a deterministic module. An LLM restricted to emitting a program, executed by a separate interpreter, beats much larger pure-LLM baselines on thirteen math and symbolic reasoning tasks including GSM8K [**?**]. Schick et al. (2023) push this further, training a model to decide on its own when to call a calculator or search tool, with large arithmetic gains while the LLM still only handles language [**?**]. Ye et al. (2023) go one step beyond that: the

LLM emits a declarative specification, an automated theorem prover checks it, and the LLM never works through the reasoning itself, improving on prior program-aided methods by 23% on a subset of GSM arithmetic problems [?]. This is the general shape of neurosymbolic AI – symbolic structures with defined semantics sitting directly in the inference loop alongside a neural component [?] – and it's also how the Hybrid architecture here is built: the LLM only extracts structured engineering parameters (mapping "a 20-year-old house with poor insulation" to an R-value and SEER estimate), and a deterministic physics-based calculator does the actual MAVT scoring. Scoring instead of solving doesn't dodge the problem, either. Li et al. (2026) document systematic scoring-bias distortions in LLM-as-a-judge pipelines that persist even in advanced models [?], and Zhu et al. (2026) trace most of the cross-run score variance on benchmarks like MT-Bench to judge identity rather than response quality [?]. The pattern isn't confined to judge benchmarks: Johnson and Zhang (2024) find GPT-4o essay scores clustering well below human ratings with noticeably less variance – central-tendency bias, where the model avoids extreme scores [?] – and Poličar et al. (2025) find that different LLM graders carry distinct, model-specific leniency or strictness profiles relative to human TAs in a bioinformatics course [?]. For Pure Prompting specifically, this predicts a clean failure mode: cluster the four criterion scores for all three alternatives into a narrow middle band, and the sharp, non-linear penalties in the ground-truth value functions – the shower calculator's tank-capacity cutoff, the budget penalty's exponential tier – get missed entirely, dragging down Kendall's tau even when the scores are directionally reasonable.

3. Methodology

3.1. Decision Types and Scenario Corpus

Three decision types were selected for their high behavioral plasticity and RAER: HVAC (thermostat setpoints), Appliance (time-of-use scheduling), and Shower (duration).

The scenario corpus has two disjoint pools. The *test set* contains 195 scenarios (70 HVAC, 65 Appliance, and 60 Shower); each scenario presents three alternatives scored independently by each architecture and by the ground-truth calculator. The *RAG corpus* contains an additional 90 scenarios (35 HVAC, 35 Appliance, and 20 Shower) that populate the retrieval index used by the RAG-Enhanced architecture; RAG scenarios are never used for evaluation [25].

[PLACEHOLDER NOTE] Justify the scenario counts and splits: 195 test (70 HVAC / 65 Appliance / 60 Shower) and 90 RAG (35 HVAC / 35 Appliance / 20 Shower). State the rationale for the per-decision-type allocation and the disjoint test/RAG partition.

The tables below detail the parameters given to architectures vs. calculators per decision type.

Table 1: HVAC parameters: Architectures vs. calculator

Parameter	**LLMs**	**Calc.**
Location	✓	—
Outdoor Temp	✓	✓
Home Size (sq ft)	✓	✓
Insulation	✓	—
R-Value	—	✓
Household Size	✓	✓
Housing Type	✓	✓
House Age[a]	✓	✓
Utility Budget	✓	✓
SEER Rating	—	✓
Occupancy Context	—	✓
Indoor Temp (from alternative)	—	✓

[a]LLMs receive a building-age range label (e.g., "20+ Years"); the calculator uses HVAC equipment age in exact years — distinct concepts.

The calculator receives specific values that must be professionally graded for households, and thus are unreasonable to be requested of homeowners. The architectures receive generalized version (e.g. insulation, which is a general

Table 2: Appliance parameters: Architectures vs. calculator. A ✓ in the LLMs column denotes a field explicitly present in the prompt; appliance type and baseline time are conveyed implicitly through the natural-language question text (“run the dryer,” “it’s around 7 in the evening”) rather than as labeled fields. TOU period, utility rate, kWh/cycle, cycle duration, and monthly-cycles are calculator-internal and are not shown to the architectures.

Parameter	LLMs	Calc.
Appliance Type	(text)	✓
Baseline / Run Time	(text)	✓
Appliance Age Range	✓	—
kWh per Cycle	—	✓
TOU Period / Utility Rate	—	✓
Monthly Cycles Assumption	—	✓
Household Size	✓	✓
Housing Type	✓	✓
Location	✓	✓
Utility Budget	✓	✓

Table 3: Shower parameters: Architectures vs. calculator.

Parameter	LLMs	Calc.
Location	✓	—
Outdoor Temp	✓	✓
Inlet Water Temp	—	✓
Heater Setpoint	—	✓
Flow Rate (low_flow/standard/high_flow)	✓	—
GPM	—	✓
Duration (min)	(alt.)	✓
Household Size	✓	✓
Tank Size	—	✓
Utility Budget	✓	✓
Housing Type	✓	✓

Duration is the alternative under evaluation (one of the three the architecture scores), not a separate context field; heater setpoint, tank size, GPM, and inlet temperature are calculator-internal and withheld from the architectures.

estimate of poor/average/good rather than a specific R-value) or do not receive counterparts to some parameters at all (e.g. occupancy context, which the architecture instead has to infer from the question text). These parameters are chosen as they are things a household member would actually know [26].

To generalize parameters, we identified acceptable ranges for each qualitative level from published standards and mapped each scenario to a representative value within that range. Insulation tiers map to standard U.S. residential wall R-values: Poor to R-11 (minimal 2×4 cavity fill, below current code), Medium to R-13 (standard 2×4 batt), and Good to R-19 (2×6 framing), consistent with code tables for wood-frame assemblies and ENERGY STAR guidance [27, 28]. For the Flow Rate parameter, the scenario’s numeric GPM is bucketed into three qualitative labels: `low_flow` (≤2.0 GPM, including EPA WaterSense certified 1.5 GPM showerheads [29]), `standard` (2.5–3.0 GPM, at or near the Energy Policy Act of 1992 federal maximum of 2.5 GPM [29]), and `high_flow` (>3.0 GPM). The label is shown to the LLM; the calculator always receives the exact numeric GPM from the scenario. Table 4 summarizes the generalized parameters across all three decision types. The two sets overlap on environmental conditions, however so that both pipelines are reasoning about the same physical scenario. The LLM never sees the engineering parameters directly. For the Hybrid architecture, which needs to use those parameters within its calculations, it infers structured estimates of them from the description before the calculator is invoked. Further detail on choosing these specific parameters and their effects is available in section 2.2.

Table 4: Parameters generalized from engineering values to homeowner-accessible labels. LLMs receive the label; calculators receive the numeric value. R-values in h·ft^2 · °F/BTU; GPM dataset range 1.5–4.0; kWh/cycle ranges span the full dataset.

Parameter	LLM Label	Calc. Value	Source
HVAC			
Insulation	Poor	R-11	[27, 28]
	Medium	R-13	
	Good	R-19	
Shower			
Flow Rate	`low_flow` (≤2.0 GPM)	scenario GPM	[29]
	`standard` (2.5–3.0 GPM)	scenario GPM	[29]
	`high_flow` (>3.0 GPM)	scenario GPM	
Appliance			
Appliance Age	1–15 yr	Dishwasher: 0.72–1.70 kWh/cyc	[30, 31, 32]
		Washer: 0.15–0.45 kWh/cyc	
		Dryer: 1.15–3.50 kWh/cyc	

3.2. MAVT Framework Design

The Multi-Attribute Value Theorem (MAVT) allows alternatives to be compared on different criterion based on the range of values that occurs for each criterion. In our study, we chose the following weights for our 4 weights: Environmental Impact (35%), Energy Cost (30%), Comfort (20%), and Practicality (15%). The weights are held constant across all decision types, all architectures, and the ground-truth calculator. We did this so that any difference in score is because of the inputs and value function, not because of difference in the way each criterion is perceived for the three decision types. Normally, MAVT additive form (Eq. 1) requires preferential independence among criteria [33, 34]. This means that criteria should be independent, meaning that the value (good or bad) of one criterion should have no effect on the value of another criterion.

$$s_j = \sum_{i=1}^{4} w_i \cdot v_i(x_{ij}) \tag{1}$$

In the case of our criteria, the energy cost and the environmental impact criterion are not completely independent, as they both stem from the kWh usage of a certain activity. The spearman's rho is *[value pending corrected metrics run]*, which does indeed mean that they are correlated. However, this is a feature of the dataset and of household emissions, and it is actually possible for the two criterion to diverge. Since environmental impact comes from actual CO_2, which varies based on the type of power plant generating the power, grid mix composition, renewable penetration, and regional interconnection flows [35], and energy cost comes from the time of day, gas peakers, demand charges, transmission congestion pricing, and utility revenue requirements [36]. While Pennsylvania does not usually have such divergence, there are other areas that do, such as Texas (ERCOT), where high daytime wind generation can drive electricity prices near zero while gas-fired backup units sustain elevated marginal emissions during evening demand peaks [37, 38].

A sensitivity analysis was conducted to test the stability of the results. Fluctuations of ±0.05 were applied to the weight vector by perturbing each of the four criterion weights individually and redistributing the difference equally across the remaining three, plus one equal-weight scenario ($w_j = 0.25$ for all j). The results can be seen in the below table, Table 5.

The Environmental weight of 0.35 reflects two convergent considerations. Residential emissions account for a sizable share of national CO_2 emissions [14], so a household decision-support tool that does not give priority to environmental impact fails. The Value–belief–norm theory, which says that show that environmental orientation is the most consistent normative driver across pro-environmental household behaviors [39, 40, 41, 42, 43, 44, 45, 46]. Finally, anyone using such a tool obviously has environmental motivations, and entropy-based normalization of the scenario distributions assigns higher information weight to the environmental criterion, independently supporting an

Table 5: Ten weight perturbation scenarios for the sensitivity analysis. Ene = Energy Cost, Env = Environmental, Com = Comfort, Pra = Practicality. Weights rounded to four decimal places. Architecture τ values are pending the corrected metrics run.

Scenario	w_{Ene}	w_{Env}	w_{Com}	w_{Pra}
Baseline	0.3000	0.3500	0.2000	0.1500
Ene +0.05	**0.3500**	0.3333	0.1833	0.1333
Ene −0.05	**0.2500**	0.3667	0.2167	0.1667
Env +0.05	0.2833	**0.4000**	0.1833	0.1333
Env −0.05	0.3167	**0.3000**	0.2167	0.1667
Com +0.05	0.2833	0.3333	**0.2500**	0.1333
Com −0.05	0.3167	0.3667	**0.1500**	0.1667
Pra +0.05	0.2833	0.3333	0.1833	**0.2000**
Pra −0.05	0.3167	0.3667	0.2167	**0.1000**
Equal	0.2500	0.2500	0.2500	0.2500

above-average allocation [47]. The sensitivity analysis (Table 5) confirms that the architecture ordering is preserved under ±0.05 perturbations of this weight, indicating that the 0.35 choice is not a threshold effect.

Cost is the strongest independent predictor of consumer adoption. This is clear for multiple reasons. First, cost savings as shown on energy labels for consumer products were more effective than actual energy use and carbon emissions.[48]. When energy labels strip away concrete cost-savings figures, the influence of all remaining information drops sharply [49, 50]. Second, households misestimate energy use, with Attari et al. reporting that participants underestimated energy use and savings by a factor of 2.8 on average [26]. Thus, high importance must be given to savings, even if the user is automating the process. It would also be beneficial to directly display actual cost savings. Weighting Energy Cost at 0.30 keeps it slightly below environmental impact while keeping its role as the criterion households actually act on. For HVAC decisions, thermostat setpoints directly determine electricity draw at the billing rate; households shown real-time cost feedback respond by reducing peak-period consumption, confirming that cost remains the operative signal even for automated decisions [51]. For shower decisions, water heating accounts for a significant share of residential energy costs and the per-shower increment is rarely visible to users, amplifying the same underestimation bias and making cost-feedback interventions similarly effective [52, 53].

Comfort is weighted at 0.20. Thermal comfort is the dominant driver of HVAC behavior even when it conflicts with energy savings [54, 55, 56], and low-income households in particular trade comfort against cost in documented ways [57]. Anchoring comfort to ASHRAE Standard 55 [58, 59] keeps the criterion physically interpretable for HVAC and provides a 0–10 scale that generalizes (with domain-specific calibration described in Section 3.3) to appliance delay and shower duration. For appliance decisions, comfort is primarily a function of run-time delay: demand-response research finds that household willingness to accept a shift decays sharply beyond two to four hours, with dishwashers tolerated longest and dryers shortest [60, 61], and the penalty scales further with household size as dishes and laundry accumulate faster. For shower decisions, comfort peaks around the REU2016 average of 7.8 minutes [52, 53] and depends additionally on temperature adequacy, with heater setpoints below CDC Legionella thresholds carrying a comfort penalty [62].

Practicality is weighted at 0.15. A persistent finding in the heuristics literature is that any tool that adds friction relative to the existing routine sees adoption drop sharply, and heuristic decision-making remains the default for everyday energy choices [15, 16, 63, 64]. Penalizing alternatives that violate budget, system, or behavioral limits keeps the tool's recommendations in check to make sure that households would actually adopt it long term. The lower weight (relative to comfort) reflects that practicality is a constraint on what is reasonable rather than a primary preference dimension. Where comfort captures what a household would prefer given no constraints, practicality captures what is feasible and sustainable to adopt. A 15-minute shower may score near-optimally on comfort, but if four occupants share a 40-gallon tank, available hot water runs out before the third person showers — making that alternative infeasible regardless of comfort preference.

The reference ranges in Table 6 are computed as the 5th–95th percentiles of the realized ground-truth quantity

Table 6: Reference ranges (5th–95th percentiles). Shower Environmental Impact is reported in gallons of water; HVAC and Appliance in lbs CO_2.

Criterion	**HVAC**	**Appliance**	**Shower**
Energy Cost ($)	0.38–3.29	0.025–0.71	0.14–1.14
Environmental Impact	1.96–18.04	0.288–3.643	6.0–45.0
Comfort	0.0–10.0	0.0–10.0	0.0–10.0
Practicality	0.5–10.0	0.5–10.0	0.5–10.0

distributions over the scenario corpus (cost and emissions for HVAC and Appliance; cost and water volume for Shower), rather than extreme theoretical limits, so that normalization preserves meaningful spread across typical residential alternatives. Using dataset percentiles rather than the full theoretical physics envelope is a deliberate modeling choice [47]: it concentrates score sensitivity in the range households actually encounter, at the cost of truncating scores for extreme outlier scenarios (documented as a paper limitation in the Discussion). The endpoints below are the percentile values to two decimals; the corresponding literature anchors are given for interpretability.

For HVAC, the cost percentiles are computed over the active-conditioning alternatives only: a setpoint at or near the outdoor temperature (and the bare "Off" option) produces zero decision-period load and collapses to $0, which would make the lower bound degenerate. Excluding those zero-load points, the realized 8-hour cost distribution at Pennsylvania's residential rate ($0.19/kWh) [7, 65] spans $0.38–$3.29 (5th–95th percentile); an Off or zero-load alternative still receives the top of the cost score via the decreasing-value normalization. These endpoints remain consistent with residential HVAC efficiency studies (efficient: [66]; degraded: [67]). The HVAC environmental bounds (1.96–18.04 lbs CO_2) are obtained by mapping the same cost-derived kWh envelope through PJM marginal emissions factors (0.976 off-peak, 1.041 peak lbs CO_2/kWh) [35], maintaining a physically consistent link between cost and emissions.

For appliances, the realized per-cycle cost distribution (kWh/cycle multiplied by the resolved time-of-use rate across the six PA utilities) spans $0.025–$0.71 at the 5th–95th percentile, and the realized per-cycle emissions distribution (kWh/cycle through the PJM marginal factor at run time) spans 0.288–3.643 lbs CO_2. The underlying per-cycle energy endpoints remain consistent with the ENERGY STAR certified-products distribution (efficient HE washers near 0.25 kWh) through older non-certified electric resistance dryers [68, 69, 70, 71, 72, 73].

For showers, the realized per-shower cost distribution spans $0.14–$1.14 at the 5th–95th percentile. Shower energy follows the mixing-fraction physics targeting a fixed 105 °F delivery temperature, so it depends only on the rise from the mains inlet (seasonal inlet temperatures from NREL hot-water modeling [74]) to that target and is independent of heater setpoint; cost is this energy priced at the PA residential rate. Shower environmental impact is defined as water volume (gallons); the realized distribution of flow rate multiplied by duration spans 6.0–45.0 gallons at the 5th–95th percentile, with flow-rate and duration benchmarks from EPA WaterSense Appendix B [29], REU2016 [52], and The Harris Poll [53].

3.3. Ground Truth Calculators

Each ground-truth calculator consumes a scenario with three alternatives and returns four scores per alternative—Energy Cost ($), Environmental Impact, Comfort, and Practicality (the latter three on 0–10)—plus the raw physical quantities before value-function transformation. Energy cost is always in dollars per 8-hour decision period; Environmental Impact is in lbs CO_2 for HVAC and Appliance decisions and in gallons of water for Shower decisions. Full equation-by-equation derivations are in Appendix A.

For HVAC and Appliance decisions, environmental impact uses PJM marginal emissions factors:

$$\mathrm{CO}_2 = E_{\mathrm{kWh}} \cdot \varepsilon(t), \qquad \varepsilon(t) = \begin{cases} 1.041\ \mathrm{lbs/kWh} & \text{peak (7am–11pm)} \\ 0.976\ \mathrm{lbs/kWh} & \text{off-peak} \end{cases} \tag{2}$$

where E_{kWh} is the decision-period energy use and $\varepsilon(t)$ is the PJM marginal emissions factor at run time t [35]. Marginal rather than average factors are used because shifting a residential load displaces or adds the generator at the margin. All three calculators apply an identical four-tier budget penalty $P(u)$ multiplicatively to the energy-cost value-function output, where $u = C_{\text{monthly}}/B$ is monthly cost utilization relative to the household's stated budget B:

$$P(u) = \begin{cases} 1.0 & u < 0.80 \\ 1 - 2.5\,(u - 0.80) & 0.80 \le u < 1.0 \\ 0.5\,e^{-3(u-1.0)} & 1.0 \le u < 1.5 \\ 0 & u \ge 1.5 \end{cases} \tag{3}$$

where $P(u)$ is a unitless multiplicative penalty on the (post value-function) energy-cost score, u is monthly budget utilization, C_{monthly} is estimated monthly energy/water-related cost for the decision type, B is the household-stated monthly budget for utilities, and e is Euler's number. The no-penalty zone ($u < 0.80$) reflects mental budget safety margins [75]; the linear decline approaching the limit follows Heath & Soll's budget self-control model [76]; exponential loss aversion for overruns follows Prelec & Loewenstein [77] and Heutel [78]; and elimination beyond 150% reflects financial infeasibility [79]. Applying the penalty multiplicatively ensures budget-infeasible alternatives cannot dominate MAVT ranking on energy cost alone.

3.3.1. HVAC Calculator

Thermal load decomposes into four ASHRAE-style components. For cooling:

$$Q_{\text{cool}} = \underbrace{\frac{m \cdot A}{R}\,\Delta T}_{\text{conductive}} + \underbrace{400n + A + 800}_{\text{internal}} + \underbrace{0.15\,A \times 20}_{\text{solar}} + \underbrace{1.08\,\frac{A\,h_c\,\alpha}{60}\,\Delta T}_{\text{infiltration}} \tag{4}$$

where A is floor area (sq ft); m is a housing-type envelope multiplier (1.7 single-family, 1.6 twin/semi-detached, 1.5 townhouse/rowhouse, 1.2 apartment/condo) [80]; R is the wall insulation R-value; $\Delta T = T_{\text{out}} - T_{\text{in}}$; n is household size; $h_c = 8$ ft and $\alpha = 0.35$ ACH for ceiling height and infiltration rate [80]. The 800 BTU/hr baseline captures always-on standby loads per ACCA Manual J [80]. The heating load uses the same expression with $\Delta T = T_{\text{in}} - T_{\text{out}}$, no solar term, and internal gains subtracted rather than added.

Effective EER is estimated from the unit's rated SEER via the AHRI 210/240 quadratic [81]:

$$\text{EER} = -0.02\,\text{SEER}^2 + 1.12\,\text{SEER} \tag{5}$$

where SEER is the unit's rated seasonal energy efficiency ratio and EER is the resulting effective energy efficiency ratio used for the decision period. Age- and maintenance-driven efficiency degradation is deliberately *not* applied to the energy path; it is instead modeled as a reliability factor in the practicality score (Eq. 9), so the energy score reflects the setpoint choice rather than the system's condition. Energy consumption over the 8-hour decision window is then:

$$E_{\text{kWh}} = \frac{Q_{\text{load}}}{\text{EER} \times 1000} \cdot h \cdot m_{\text{occ}} \tag{6}$$

where Q_{load} is the thermal load (BTU/hr) for the decision context, h is the decision-window duration (8 hr), and m_{occ} is an occupancy multiplier. where $h = 8$ hr and m_{occ} scales for occupancy context: 1.0 for fully occupied, 0.75 for overnight, and $1 - 0.5(h_{\text{away}}/24)$ for daytime unoccupied periods [82], where h_{away} is hours-away per 24-hour day.

Because HVAC alternatives carry no explicit run-time, the marginal emissions factor in Eq. 2 is resolved from the occupancy context by mapping each pattern onto the PJM peak (7am–11pm, 16 h) and off-peak (8 h) windows, with away-hours assumed to fill the daytime peak window first:

$$\varepsilon_{\text{occ}} = \begin{cases} \varepsilon_{\text{off}} & \text{occupied-sleep (night-only runtime)} \\ \dfrac{16\,\varepsilon_{\text{pk}} + 8\,\varepsilon_{\text{off}}}{24} & \text{occupied all day} \\ \varepsilon_{\text{pk}} & \text{unoccupied } H \text{ hr},\ H \le 16 \\ \dfrac{16\,\varepsilon_{\text{pk}} + (H-16)\,\varepsilon_{\text{off}}}{H} & \text{unoccupied } H \text{ hr},\ H > 16 \end{cases} \tag{7}$$

where $\varepsilon_{\text{pk}} = 1.041$ and $\varepsilon_{\text{off}} = 0.976$ lbs/kWh are the PJM peak and off-peak marginal factors and H is hours-away per day [35]. Occupied-sleep returns the off-peak factor because the home's conditioning runtime falls entirely in the overnight off-peak window.

Comfort uses a tent function anchored to ASHRAE 55 PMV-optimal setpoints [83, 84, 85]:

$$C_{\text{comfort}} = \max\left(0,\ 10 - |T_{\text{in}} - T^*|\right) \tag{8}$$

where C_{comfort} is a 0–10 comfort score, T_{in} is the indoor setpoint, and T^* is the context-dependent optimal setpoint used as the peak of the tent function. where $T^* = 76\,°\text{F}$ in cooling mode ($T_{\text{out}} > 75\,°\text{F}$) and $70\,°\text{F}$ in heating mode, consistent with ASHRAE 55-2020 sedentary PMV bands [83]. A small penalty scales with household size beyond three occupants. Practicality starts at 10 and subtracts an asymmetric extremity penalty for setpoints beyond adoption-comfort bounds (in cooling, $\geq 82\,°\text{F}$ or $\leq 71\,°\text{F}$; in heating, $\leq 63\,°\text{F}$ or $\geq 76\,°\text{F}$), with the steepest per-°F slopes on the least-tolerable directions (1.5/1.0 for the high/low side in cooling; 1.8/0.8 for the low/high side in heating). The result is then multiplied by an outdoor–indoor ΔT feasibility factor (1.0, 0.95, 0.85, 0.70 for $\Delta T < 10, < 20, < 35, \geq 35\,°\text{F}$), signaling operation near system limits. Finally, age- and maintenance-driven efficiency degradation enters here as a reliability factor (item relocated from the energy path): an older or poorly maintained unit is a less practical choice to rely on, so the score is scaled by $(1 - D)$ where

$$D = \min\left(0.30,\ \rho(\text{maint})\left[\,1.5\min(a, 10) + 0.5\max(a - 10, 0)\,\right]\right) \tag{9}$$

is the fraction of efficiency lost (capped at 30%), a is HVAC age in years, and $\rho(\text{maint})$ is the base annual loss rate (0.005, 0.010, 0.015 per year for good, moderate, and poor upkeep); the first ten years degrade at 1.5ρ and later years at 0.5ρ [86, 87, 88]. The final practicality score is clipped to [1.5, 10].

An "Off" alternative (no active conditioning) is scored with zero decision-period energy, cost, and emissions; its comfort and practicality are instead evaluated against the free-float indoor temperature the building drifts to with the system off — outdoor +5 °F in cooling season and +10 °F in heating season, reflecting solar and internal gains that hold a closed, occupied house above outdoor air temperature [59, 83, 89, 90].

3.3.2. Appliance Calculator

Energy cost per cycle is $C = E_{\text{cyc}} \cdot r(t, \ell)$, where E_{cyc} is kWh/cycle and $r(t, \ell)$ is the location- and time-specific TOU rate resolved to one of six Pennsylvania utilities (PECO, PPL, West Penn, Penelec, MetEd, Duquesne) [36]. where C is per-cycle energy cost (dollars/cycle), E_{cyc} is per-cycle energy use (kWh/cycle), t is the appliance run time (time-of-day), and ℓ is the household location used to resolve the serving utility. The PJM emissions window (7am–11pm) applies to environmental impact regardless of the utility's billing window, because emissions follow the regional grid. The run-time rate period is selected by mapping location ℓ to a utility $U(\ell)$ and then applying that utility's TOU peak window otherwise.

$$(h^U_{\text{pk,s}}, h^U_{\text{pk,e}}): \quad r(t, \ell) = r^U_{\text{pk}} \text{ if } h(t) \in [h^U_{\text{pk,s}}, h^U_{\text{pk,e}}) \text{ and } r(t, \ell) = r^U_{\text{off}} \tag{10}$$

where $U(\ell)$ maps location ℓ to a serving utility, $h^U_{\text{pk,s}}$ and $h^U_{\text{pk,e}}$ are that utility's TOU peak-window start and end hours, r^U_{pk} and r^U_{off} are the utility's peak and off-peak energy prices ($/kWh), and $h(t)$ maps time t to an hour-of-day on a 24-hour clock. In the implementation, the six utilities use distinct peak windows (e.g., 2–6pm for PECO and PPL, 2–9pm for most FirstEnergy utilities).

[PLACEHOLDER NOTE] Weekend and holiday peak-hour pricing not currently modeled; all days treated as weekdays for TOU rate assignment.

[PLACEHOLDER NOTE] Seasonal peak-window shifts not currently modeled: PPL's winter (Dec–May) peak window is 4–8pm versus the 2–6pm summer window applied here; the summer window is used year-round because scenarios carry no season or date field. Comfort is a piecewise decay from 10 at zero delay, with appliance-specific tolerance ceilings of 12 hr (dishwashers), 8 hr (washers), and 6 hr (dryers) [60, 61], plus an additive noise penalty scaled by housing type for runs after 10 pm [91] and a household-size penalty. Delays are computed as the minimum circular distance from the baseline time t_0 on a 24-hour clock (avoiding wrap-around artifacts), i.e.,

$$d(t, t_0) = \min\left((h(t) - h(t_0)) \bmod 24,\ (h(t_0) - h(t)) \bmod 24\right), \tag{11}$$

where $d(t, t_0)$ is delay in hours between run time t and baseline time t_0, $h(\cdot)$ maps a time to hour-of-day, and mod24 indicates wrap-around on a 24-hour clock. The base comfort component uses the following piecewise schedule on delay hours d:

$$C_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - \frac{2}{3}d & 0 < d \le 3 \\ 8 - \frac{1}{2}(d-3) & 3 < d \le 7 \\ 6 - \frac{2}{5}(d-7) & 7 < d \le 12 \\ 2 & d > 12 \end{cases} \tag{12}$$

where $C_{\text{base}}(d)$ is the pre-penalty comfort score (0–10) as a function of delay d. The late-night noise penalty is applied when running between 10pm and 7am and the appliance exceeds a dBA threshold:

$$N(t, \text{type}, \text{housing}) = \begin{cases} 0 & \text{if } h(t) \in [7, 22) \text{ or dBA(type)} \le \theta \\ 2 \cdot \kappa(\text{housing}) & \text{otherwise} \end{cases} \tag{13}$$

where $N(t, \text{type}, \text{housing})$ is a nonnegative noise-disruption penalty in comfort-score points, dBA(type) is the appliance noise level for the appliance type, θ is the dBA threshold, and κ(housing) is a unitless housing-type multiplier (larger for shared-wall housing). where $\theta = 45\,\text{dBA}$ and $\kappa(\cdot)$ is a housing-type multiplier (larger for shared-wall housing). The household-size penalty scales with delay relative to the appliance-specific maximum acceptable delay $d_{\max}$(type):

$$S(d, n, \text{type}) = \sigma(n) \cdot \min\left(\frac{d}{d_{\max}(\text{type})}, 1\right), \tag{14}$$

where $S(d, n, \text{type})$ is a nonnegative household-size penalty in comfort-score points, n is occupants, $d_{\max}$(type) is the maximum acceptable delay (hours) for the appliance type, and $\sigma(n)$ is an occupants-based factor (higher for larger households). where $\sigma(n)$ is an occupants-based factor (higher for larger households). Final comfort is then $C_{\text{comfort}} = \text{clip}_{[0,10]}(C_{\text{base}} - N - S)$. where $\text{clip}_{[a,b]}(x) = \min(b, \max(a, x))$ clamps a value to the interval $[a, b]$. Practicality mirrors the delay schedule with additional penalties for late-night timing complexity and household coordination difficulty. The base practicality component is:

$$P_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - d & 0 < d \le 2 \\ 8 - \frac{3}{4}(d-2) & 2 < d \le 4 \\ 6.5 - \frac{1}{2}(d-4) & 4 < d \le 8 \\ 4.5 - \frac{3}{8}(d-8) & 8 < d \le 12 \\ 1.5 & d > 12 \end{cases} \tag{15}$$

where $P_{\text{base}}(d)$ is the pre-penalty practicality score (0–10) as a function of delay d. Timing complexity applies an additive penalty by run-time hour:

$$T(t) = \begin{cases} 2 & 0 \le h(t) < 6 \\ 1 & 22 \le h(t) < 24 \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

where $T(t)$ is a nonnegative timing-complexity penalty in practicality-score points and $h(t)$ is hour-of-day. Household coordination difficulty uses the same delay scaling form as in comfort:

$$K(d, n, \text{type}) = \kappa_n(n) \cdot \min\left(\frac{d}{d_{\max}(\text{type})}, 1\right), \tag{17}$$

where $K(d, n, \text{type})$ is a nonnegative household-coordination penalty in practicality-score points and $\kappa_n(n)$ is an occupants-based factor increasing with occupants. with $\kappa_n(n)$ increasing with occupants. Final practicality is $C_{\text{prac}} = \text{clip}_{[1.5,10]}\big(P_{\text{base}} - T - K\big)$, with a daytime floor applied for $h(t) \in [7, 22)$ and $d \leq d_{\max}(\text{type})$. where C_{prac} is the final 0–10 practicality score (after penalties and clipping). For budget utilization in (3), the appliance calculator converts per-cycle cost to a monthly estimate as:

$$C_{\text{monthly}} = c(\text{type})\, C_{\text{cycle}}, \qquad c(\text{type}) = \begin{cases} 18 & \text{dishwasher} \\ 25 & \text{washer} \\ 24 & \text{dryer} \end{cases} \tag{18}$$

where C_{cycle} is per-cycle energy cost and $c(\text{type})$ is the appliance-specific cycles-per-month estimate, from 215, 300, and 283 cycles/year for dishwashers, washers, and dryers respectively [30, 92, 93].

3.3.3. Shower Calculator.

Mains inlet temperature is interpolated from outdoor temperature between 45 °F (winter, $T_{\text{out}} \leq 32$ °F) and 65 °F (summer, $T_{\text{out}} \geq 75$ °F) [74, 94]. Only the fraction of flow requiring heating reaches the heater; that mixing fraction is:

$$f_{\text{hot}} = \frac{T_{\text{target}} - T_{\text{inlet}}}{T_{\text{heater}} - T_{\text{inlet}}} \tag{19}$$

where f_{hot} is the fraction of total flow drawn as hot water, T_{target} is desired delivery temperature, T_{inlet} is mains inlet temperature, and T_{heater} is the water-heater setpoint temperature. where $T_{\text{target}} = 105$ °F is the standard comfortable delivery temperature [95]. Shower energy is then:

$$E_{\text{kWh}} = \frac{\dot{V} \cdot f_{\text{hot}} \cdot 8.33 \cdot (T_{\text{heater}} - T_{\text{inlet}}) \cdot \tau}{3412 \cdot \eta} \tag{20}$$

where E_{kWh} is shower energy use (kWh), $\dot{V}$ is total flow rate (GPM), τ is shower duration (min), η is electric heater efficiency (UEF), and 3412 is the BTU-per-kWh conversion; the constant 8.33 converts gallons to pounds of water (lb/gal). where $\dot{V}$ is flow rate (GPM), 8.33 lb/gal is water density, τ is shower duration (min), 3412 BTU/kWh is the standard conversion, and $\eta = 0.92$ is the unified energy factor for a 40–55 gal electric tank [96]. Environmental impact is water volume: $V = \dot{V} \cdot \tau$ (gallons). where V is water volume used (gallons). Energy cost uses a flat electricity rate p_e:

$$C_{\$} = p_e\, E_{\text{kWh}}. \tag{21}$$

where $C_{\$}$ is shower energy cost (dollars) and p_e is the electricity price ($/kWh). Comfort is a piecewise function peaking at the REU2016 average of 7.8 min [52, 53], with additive penalties for temperature adequacy (CDC Legionella thresholds [62]) and household contention. The implementation uses a duration-based piecewise base comfort $C_{\text{base}}(\tau)$ (with a short-duration cubic ramp to reflect severely rushed showers), then subtracts a temperature penalty $\Delta_{\text{temp}}(T_{\text{heater}})$ and a household contention penalty $\Delta_{\text{cont}}(\tau, n)$, and clips to $[0, 10]$. The comfort-optimal duration is the 7.8 min REU2016 mean at moderate outdoor temperatures but is shifted upward in cold weather (to at most $\approx$10.3 min, the observed winter field-mean envelope), reflecting that comfortable showering lengthens as it gets colder; this cold-weather elasticity is an explicit modeling assumption supported only in direction by the literature [52, 53], and warm-weather scoring is unchanged:

$$C_{\text{comfort}}(\tau, T_{\text{heater}}, n) = \text{clip}_{[0,10]}\Big(C_{\text{base}}(\tau) - \Delta_{\text{temp}}(T_{\text{heater}}) - \Delta_{\text{cont}}(\tau, n)\Big). \tag{22}$$

where τ is shower duration (min), n is occupants, $C_{\text{base}}(\tau)$ is the base comfort curve, $\Delta_{\text{temp}}(T_{\text{heater}})$ is a temperature-adequacy penalty, and $\Delta_{\text{cont}}(\tau, n)$ is a household-contention penalty. Practicality additionally penalizes alternatives that exceed available tank capacity. Unlike comfort, the base practicality curve is not centered on the 7.8 min comfort optimum: it rises with duration up to a peak near 12 min before declining, because practicality here measures *adoptability*—how little behavior change an alternative demands of a household that already showers at a given length—rather than momentary preference. A very short shower scores low on practicality (it requires sustained restraint), while a moderately long one is easy to sustain. Pressure toward shorter showers in the MAVT aggregate therefore comes from the cost and environmental criteria and from the comfort optimum, not from practicality; the capacity constraint below is what makes an over-long shower infeasible for a constrained household. The practicality model uses a duration-based base practicality $P_{\text{base}}(\tau)$ and a capacity constraint computed from hot-water demand. Hot-water per shower is $\tau \dot{V} f_{\text{hot}}$ (gal), total household hot-water need is $n \tau \dot{V} f_{\text{hot}}$, and available draw is approximated as 0.8 of nominal tank size S:

$$H_{\text{need}} = n \tau \dot{V} f_{\text{hot}}, \qquad H_{\text{avail}} = 0.8\, S. \tag{23}$$

where H_{need} is total hot-water volume required for the household's showers over the relevant period (gal), H_{avail} is available hot-water draw volume (gal), and S is nominal tank capacity (gal). A fixed penalty is applied when $H_{\text{need}} > H_{\text{avail}}$, yielding a clipped final practicality score. For budget utilization in (3), shower monthly cost is estimated from per-shower cost using an average showers-per-person-per-day rate:

$$C_{\text{monthly}} = C_{\text{shower}} \cdot (n \cdot 0.9 \cdot 30). \tag{24}$$

where C_{shower} is per-shower energy cost, n is occupants, 0.9 is showers/person/day, and 30 is days/month.

3.4. Architecture Implementations

All three architectures call the same underlying model via OpenRouter at temperature 0.3, with model held constant within each benchmark run so that architecture differences rather than model differences drive the comparison. The study evaluates four models: GPT-OSS-20B (reasoning, low effort), Qwen 3.5 9B (non-reasoning), DeepSeek V4 Flash (non-reasoning), and Gemini 3.5 Flash (reasoning, minimal effort). *[BLOCKED: update model descriptions and results after final multi-model runs.]* Each architecture is described below; pipeline schematics appear in Figure 4.

3.4.1. Pure Prompting

The Pure Prompting architecture issues one API call per alternative (three calls per scenario) using a calibrated system prompt that combines role-prompting [97], structured-output constraints [98], and explicit bias-mitigation language drawn from Galaz et al. on AI sustainability priors [99]. The model is asked to return JSON with all four criterion scores on a 0–10 scale; no external context is supplied beyond the scenario description.

3.4.2. RAG-Enhanced

The RAG-Enhanced architecture retrieves the top $k = 3$ most similar scenarios from a ChromaDB vector store before scoring, using the sentence-transformers/all-MiniLM-L6-v2 embedding model. Each retrieved exemplar is rendered as a full worked example: its homeowner-facing parameters, its withheld engineering parameters (R-value, SEER, GPM, kWh/cycle, tank size, heater setpoint, etc.), and each of its alternatives with the four ground-truth criterion scores plus the MAVT aggregate and rank. This is deliberate—the exemplar's engineering values are shown so the model can see how those values map to scores—and it does not leak the *target* scenario's withheld values, which are never placed in the prompt. The model is instructed to use the exemplars as reference but to score the target scenario independently. Like Pure Prompting, RAG-Enhanced uses three API calls per scenario (one per alternative). The retrieval index is built once from the 90 RAG-only scenarios described in Section 2.1 and is never queried with test scenarios [25].

3.4.3. Hybrid

The Hybrid architecture issues a single API call per scenario for structured parameter extraction. Known homeowner-visible scenario fields (outdoor temperature, square footage, insulation tier, household size, housing type, utility budget, and location) are passed directly to the calculator without LLM involvement, and the three alternatives are taken verbatim from the scenario sheet—the deterministic calculator, not the LLM, parses the numeric content of each alternative string (setpoint temperature, run time, or shower duration). The LLM is asked only to estimate the genuinely withheld engineering parameters that are absent from the homeowner-visible description: for HVAC, the wall R-value, SEER rating, HVAC age, and occupancy context; for Appliance, the appliance type, kWh/cycle, and current baseline time; for Shower, the numeric GPM (estimated from the homeowner-facing flow-rate label), tank size, and water-heater setpoint. The extracted JSON is validated—each numeric parameter must parse as a finite value within its physically admissible range, and the extracted decision type must match the scenario's known type, or the scenario is recorded as an extraction failure—and then merged with the known fields and passed to the corresponding ground-truth calculator. Because the alternatives and all homeowner facts are supplied deterministically and only the scenario-level engineering parameters are model-estimated, the LLM's output cannot inject a per-alternative scoring artifact; the calculator does the physics identically to the ground truth. Hybrid uses 1 API call per scenario versus 3 for Pure and RAG, exploiting decomposed prompting [100] to separate natural-language understanding from quantitative computation.

3.5. Evaluation Metrics

Accuracy is evaluated on two axes. Score-level error uses Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) between architecture scores and ground-truth scores on the 0–10 MAVT scale, computed separately for each criterion and aggregated across criteria. Ranking accuracy uses Kendall's τ and Spearman's ρ between the architecture's induced ranking of the three alternatives and the ground-truth ranking, together with Top-1 accuracy (architecture's best-ranked alternative matches ground truth) and Top-2 accuracy (ground-truth top alternative appears in the architecture's top two). Efficiency is evaluated using total tokens (input plus output) and total latency, both summed across the 195-scenario test set.

Failure is recorded per scenario via a reserved sentinel value (1928) that marks any malformed output, missing score, out-of-bounds score, or invalid score type, and—for Hybrid—any failed parameter extraction (non-JSON output, an invalid or mismatched decision type, or a numeric engineering parameter that does not parse as a finite value in its physical range) or ground-truth calculation error. The accuracy metrics above are therefore *conditioned on success*: a scenario containing any sentinel is excluded from the MAE/RMSE and ranking computations and is instead counted in a separately reported failure rate, so the two are read together (e.g., "$\tau = x$ over the $1 - f$ fraction that succeeded"). The architectures differ in failure granularity, which we note when comparing rates: Pure Prompting and RAG-Enhanced issue one call per alternative and mark the whole scenario failed if any one alternative fails, whereas Hybrid issues a single extraction call per scenario, so its failure unit is inherently the scenario. Scenario matching between architectures and ground truth uses (question, location) content keys plus decision-type-specific disambiguating parameters, to handle the case where scenario IDs drift across runs.

3.6. End-to-End Workflow

Figure 3 shows the full decision-support workflow common to all three architectures: a natural-language household scenario is converted into MCDA alternatives, criteria, and contextual parameters, scored on the four criteria, and aggregated by the MAVT weighted sum into a ranked recommendation. The three architectures differ only in how the scoring step is performed.

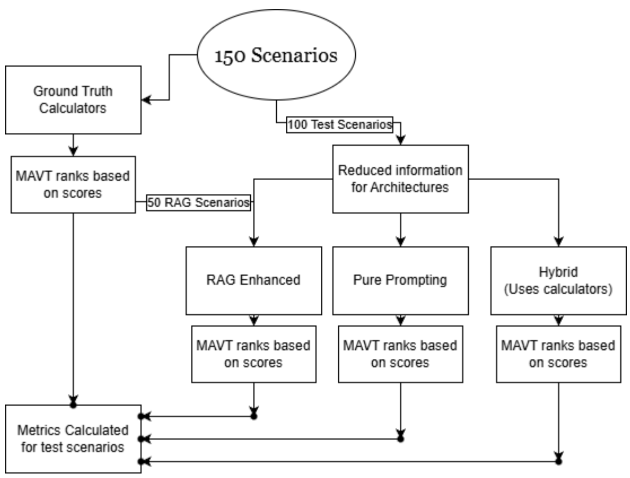

Figure 3: End-to-end MCDA decision-support workflow shared by the three architectures, from natural-language scenario to MAVT-ranked recommendation.

4. Results

[NOTE: All numeric values in this section are placeholders pending the corrected multi-model benchmark run. The subsection scaffolding, table structure, and interpretation logic below are final; replace each `XX`/`[...]` token with the value produced by `CalculateMetrics.py` aggregated over the $N = 10$ runs per model. Report every accuracy metric conditioned on success *alongside the failure rate from Section 3.5.]*

4.1. Overall Ranking Accuracy

Table 10 reports overall ranking accuracy (Kendall's τ, Spearman's ρ, Top-1, Top-2) and score-level error (MAE, RMSE) for the three architectures, averaged over the $N = 10$ runs per model and pooled across the 195 test scenarios. *[CONFIRM AFTER RUNNING: state the architecture ordering on τ and Top-1, the size of the Hybrid>RAG and RAG>Pure gaps, and whether the ordering is consistent across all four models. Because the test scenarios are shared across architectures, evaluate pairwise differences with a paired test (e.g. Wilcoxon signed-rank over per-scenario τ) rather than an unpaired comparison, and report the p-value and effect size.]* Per the structural argument in Section 5.1, the ranking gap is interpreted jointly with the score-level error in Section 4.2: ranking robustness alone is partly guaranteed by the scenario-level nature of the extracted parameters, so the score-level error is the stronger evidence of extraction fidelity.

4.2. Criterion-Level Error

Table 11 breaks MAE down by criterion. *[CONFIRM AFTER RUNNING: identify which criteria drive each architecture's error; note that Hybrid's comfort/practicality error reflects only the extracted scenario-level parameters (e.g. occupancy, tank size) since the alternative values are parsed deterministically, whereas Pure/RAG error reflects the model's direct numeric judgment. Report the Shower environmental MAE separately, as it is on the water-volume scale (gallons) rather than CO_2 (Section 3.3).]*

4.3. Performance by Decision Type

Table 12 reports τ and Top-1/Top-2 per decision type. *[CONFIRM AFTER RUNNING: report whether the Hybrid advantage holds within each of HVAC, Appliance, and Shower, and whether the Appliance result—the one decision type whose extracted parameter (baseline time) varies in effect across alternatives—tracks extraction quality more closely than HVAC/Shower, as the structural argument predicts.]*

4.4. Token Usage, Latency, and Failure Rate

Table 13 reports total API calls, token usage, latency, and success/failure rates. By construction Hybrid issues one call per scenario versus three for Pure and RAG. *[CONFIRM AFTER RUNNING: report the realized token and latency totals, the per-architecture failure rate (and, for Hybrid, the extraction-vs-calculation breakdown), and whether the deterministic calculator component measurably lowers run-to-run score variance relative to the two fully LLM-driven architectures (compare the criterion `*_std` columns). Restate that accuracy metrics are conditioned on the successful fraction.]*

5. Discussion

5.1. Limitations

The benchmark is run over four models spanning a range of capability and open/closed weights—GPT-OSS-20B (reasoning, low effort), Qwen 3.5 9B (non-reasoning), DeepSeek V4 Flash (non-reasoning), and Gemini 3.5 Flash (reasoning, minimal effort)—all called through OpenRouter at temperature $T = 0.3$, with the model held constant within each benchmark run so architecture differences rather than model differences drive each comparison. *[CONFIRM AFTER RUNNING: state whether architecture-level conclusions can be separated from model-level effects and whether the Hybrid advantage holds across model families.]*

A multi-run protocol of $N = 10$ runs per model is defined in the codebase and standard deviation columns are implemented in `CalculateMetrics.py`. [CONFIRM AFTER RUNNING: report observed variance across runs and assess whether performance gaps between architectures are statistically significant.] Until then, all reported metrics—Kendall's τ, Top-1 accuracy, MAE—reflect a single run at temperature $T = 0.3$ and are not accompanied by confidence intervals.

The most critical validity threat is that the same researcher designed both the ground truth calculators and the Hybrid architecture. Hybrid's advantage may partly reflect design alignment—the calculator the Hybrid calls is the same one that produces the ground truth—rather than a generalizable architectural superiority. This threat cannot be eliminated within the current study design and must be weighed when interpreting the magnitude of Hybrid's performance gap over RAG-Enhanced and Pure Prompting.

A related and more subtle point concerns *what* the Hybrid ranking metrics actually test. Every engineering parameter the Hybrid LLM estimates (HVAC: R-value, SEER, HVAC age, occupancy context; Appliance: kWh/cycle; Shower: GPM, tank size, heater setpoint) is a *scenario-level* quantity—identical across the three alternatives being ranked within a scenario. Because the calculator applies the same estimated parameter to all three alternatives, an estimation error shifts all three raw values together and largely cancels in the *within-scenario ranking*; it survives only through nonlinearities (value-function clipping, the budget-penalty thresholds, the shower tank-capacity threshold) and through the single per-alternative quantity the LLM influences, the appliance baseline time. Consequently, Hybrid's ranking accuracy (τ, Top-1/Top-2) is expected to be robust to weak extraction almost by construction, whereas the score-level error (MAE/RMSE) and the Appliance ranking are the measures that genuinely probe extraction quality. We therefore interpret a model-invariant Hybrid *ranking* advantage as partly structural rather than as evidence that weak models extract accurately, and we report the score-level error and the per-decision-type ranking separately so the two effects are not conflated. *[CONFIRM AFTER RUNNING: report a structural floor for Hybrid ranking by re-scoring with deliberately corrupted / randomized extracted parameters, so the observed τ can be compared against the rank-invariance baseline; state how far above that floor each model's Hybrid τ sits.]*

Results are specific to Pennsylvania's grid and utility rate structure. The PJM marginal emissions factors (peak: 1.041 lbs CO_2/kWh; off-peak: 0.976 lbs CO_2/kWh) and Pennsylvania time-of-use rate tiers used in the ground truth calculators do not generalize to high-renewables grids, where lower and more variable emissions intensities would compress environmental score distributions and potentially change which architecture performs best.

The environmental criterion for shower decisions uses water consumption in gallons as a proxy rather than CO_2 emissions. This makes the environmental dimension semantically heterogeneous across decision types: HVAC and Appliance environmental scores reflect grid emissions, while Shower environmental scores reflect water volume. Cross-decision-type comparisons of environmental MAE should be interpreted with this distinction in mind; the proxy does not affect within-type ranking validity.

The prompts have variations between architectures. While some of this is necessary due to their purposes, this could influence results. Pure-prompting was designed with the assistance of fuzzy anchors [CITE HERE FROM OUR CITATIONS ABT THIS], which helps the LLM understand some basic physics fundamentals and gives it a way to triangulate scores [CITE FOR TRIANGULATION]. This is not present in the other two architectures. RAG has the same body text as Pure-Prompting, however it replaces the fuzzy anchor section with a much longer section on 3 expert scored scenarios. Finally, Hybrid has a overall different structure, as it asks the LLM to extrapolate parameters from the given decision as opposed to scoring anything.

A more fundamental constraint concerns the implementation cost of the Hybrid architecture itself. The three ground-truth calculators developed for this study—covering HVAC load estimation, appliance TOU scheduling, and shower thermodynamics—each required extensive domain research, iterative calibration, and verification against published standards (ASHRAE, ENERGY STAR, EPA eGRID). In practice, this engineering effort constituted the majority of the total project time. This cost does not diminish Hybrid's accuracy advantage, but it does bound its practical applicability: Hybrid is the appropriate architecture when a vetted, domain-specific calculator already exists or is worth building for the decision at hand. For decision domains where no such calculator exists and the cost of developing one is prohibitive, RAG-Enhanced or sufficiently capable Pure Prompting may remain the only accessible options, even at a known accuracy cost. This tradeoff is the primary practical implication of RQ2 and should be weighed carefully before adopting the Hybrid approach in new domains.

5.2. Future Work

The immediate priority is completing the multi-model benchmark. The pipeline is fully configured for GPT-OSS-20B, Qwen 3.5 9B, DeepSeek V4 Flash, and Gemini 3.5 Flash, spanning open versus closed-weight and a range of small-to-large capability classes. [CONFIRM AFTER RUNNING: state whether the Hybrid advantage is robust or model-dependent, and summarize architecture-level conclusions across all four models.]

The per-run file naming infrastructure and standard deviation columns in `CalculateMetrics.py` are already implemented for $N = 10$ runs per model. [CONFIRM AFTER RUNNING: report whether Hybrid's deterministic calculator component measurably suppresses run-to-run variance relative to the two fully LLM-driven architectures.]

If the corrected runs reproduce the earlier pattern of a Hybrid ranking drop from HVAC to Appliance and Shower *[CONFIRM AFTER RUNNING: report the per-decision-type Hybrid τ and whether the HVAC-to-other gap persists]*, scenario-type-specific extraction prompts for Appliance and Shower decisions should be developed to address it, as the current unified extraction prompt was calibrated primarily against HVAC scenarios.

Replacing the water-consumption proxy in the Shower environmental criterion with a lifecycle CO_2 factor—converting gallons of hot water to grid emissions using a standardized heating efficiency—would make the environmental criterion semantically consistent across all three decision types and enable valid cross-type environmental MAE comparisons.

6. Conclusion

[NOTE: Conclusion scaffold; fill the bracketed quantitative claims from the corrected multi-model run. The framing is final.]

This study benchmarked three strategies for integrating an LLM into an MCDA workflow for household energy decisions—direct Pure Prompting, RAG-Enhanced prompting, and a Hybrid that uses the LLM only to extract withheld engineering parameters and a deterministic physics calculator to score—against a physics-based MAVT ground truth across HVAC, Appliance, and Shower decisions, with the model and weighting held constant so that the integration strategy is the variable under test. *[CONFIRM AFTER RUNNING: state the headline ordering and the magnitude of the Hybrid advantage, and whether it persists across all four models.]*

The central finding is that grounding the LLM's numeric reasoning in a deterministic calculator—rather than asking it to produce scores directly—*[CONFIRM AFTER RUNNING: improves / does not improve]* agreement with the physics ground truth, and *[CONFIRM AFTER RUNNING: does / does not]* hold as model capability decreases, supporting the argument that a physics backbone can compensate for weaker models. We deliberately separate this claim into its ranking and score-level components: the ranking robustness of the Hybrid design is partly structural (its extracted parameters are scenario-level and largely cancel within a ranking), so the score-level error and the appliance-timing result are the measures that most directly test extraction fidelity, and they should be read as the load-bearing evidence. The two principal threats to this conclusion—that one researcher designed both the calculator and the Hybrid extractor, and the single-grid (Pennsylvania/PJM) specificity of the ground truth—are stated in Section 5.1 and bound the strength of the generalization. Future work extending the benchmark across grids and adding scenario-type-specific extraction prompts would test whether the Hybrid advantage is an artifact of design alignment or a transferable property of physics-grounded LLM-MCDA.

References

[1] U. E. P. Agency, Assumptions and references for household carbon footprint calculator, Tech. rep., U.S. Environmental Protection Agency, washington, DC. (2024).

[2] W. contributors, Domestic energy consumption, section: Average domestic energy consumption per household in the United States (2024).

[3] C. M. Jones, D. M. Kammen, Quantifying carbon footprint reduction opportunities for u.s. households and communities, Environmental Science & Technology 45 (9) (2011) 4088–4095. doi:10.1021/es102221h.

[4] U. o. M. Center for Sustainable Systems, Carbon footprint factsheet, Tech. rep., Center for Sustainable Systems, University of Michigan, pub. No. CSS09-05 (2023).

[5] A. M. Ohler, D. G. Loomis, K. Ilves, A study of electricity savings from energy star appliances using household survey data, Energy Policy 144 (2020) 111607. doi:10.1016/j.enpol.2020.111607.

[6] U. D. of Energy, Why energy efficiency matters, u.S. Department of Energy webpage.

[7] U. E. I. Administration, Frequently asked questions (faqs), Tech. rep., U.S. Energy Information Administration, 2022. (2022).

[8] B. Goldstein, D. Gounaridis, J. P. Newell, The carbon footprint of household energy use in the united states, Proceedings of the National Academy of Sciences 117 (32) (2020) 19122–19130. doi:10.1073/pnas.1922205117.

[9] A. Kollmuss, J. Agyeman, Mind the gap: Why do people act environmentally and what are the barriers to pro-environmental behavior?, Environ. Educ. Res. 8 (3) (2002) 239–260. doi:10.1080/13504620220145401.

[10] W. Collinge, A. Landis, A. Jones, L. Schaefer, M. Bilec, Productivity metrics in dynamic lca for whole buildings: Using a post-occupancy evaluation of energy and indoor environmental quality tradeoffs, Build. Environ. 82 (2014) 339–348. doi:10.1016/j.buildenv.2014.08.032.

[11] I. Huang, J. Keisler, I. Linkov, Multi-criteria decision analysis in environmental sciences: Ten years of applications and trends, Sci. Total Environ. 409 (19) (2011) 3578–3594. doi:10.1016/j.scitotenv.2011.06.022.

[12] J. Cerneviciene, A. Kabasinskas, Review of multi-criteria decision-making methods in finance using explainable artificial intelligence, Front. Artif. Intell. 5 (2022). doi:10.3389/frai.2022.827584.

[13] H. Huang, P. Burgherr, Mcda calculator: A streamlined decision support system for multi-criteria decision analysis, Tech. rep., H. Huang and P. Burgherr, in: S.P. Duarte, et al. (Eds.), Decision Support Systems XIV. ICDSST 2024. Lecture Notes in Business Information Processing, vol. 506, Springer, Cham, 2024. (2024).

[14] T. Dietz, G. Gardner, J. Gilligan, P. Stern, M. Vandenbergh, Household actions can provide a behavioral wedge to rapidly reduce us carbon emissions, Proc. Natl. Acad. Sci. U.S.A. 106 (44) (2009) 18452–18456. doi:10.1073/pnas.0908738106.

[15] D. Ariely, D. Zakay, A timely account of the role of duration in decision making, Acta Psychol. 108 (2) (2001) 187–207. doi:10.1016/S0001-6918(01)00034-8.

[16] F. Leclerc, B. Schmitt, L. Dube, Waiting time and decision making: Is time like money?, J. Consum. Res. 22 (1) (1995) 110–119. doi:10.1086/209439.

[17] F.-H. Nah, A study on tolerable waiting time: How long are web users willing to wait?, Behav. Inf. Technol. 23 (3) (2004) 153–163. doi:10.1080/01449290410001669914.

[18] S. Bobadilla-Suarez, B. C. Love, Fast or frugal, but not both: Decision heuristics under time pressure, Journal of Experimental Psychology: Learning, Memory, and Cognition 44 (1) (2018) 24–33. doi:10.1037/xlm0000419.

[19] M. Hahn, R. Lawson, Y. Lee, The effects of time pressure and information load on decision quality, Psychol. Mark. 9 (5) (1992). doi:10.1002/mar.4220090503.

[20] G. Phillips-Wren, M. Adya, Decision making under stress: The role of information overload, time pressure, complexity, and uncertainty, J. Decis. Syst. 29 (sup1) (2020) 213–225. doi:10.1080/12460125.2020.1768680.

[21] I. Jackson, D. Ivanov, A. Dolgui, J. Namdar, Generative artificial intelligence in supply chain and operations management: A capability-based framework for analysis and implementation, Int. J. Prod. Res. 62 (2024) 6120–6145. doi:10.1080/00207543.2024.2309309.

[22] M. Huang, R. Rust, A strategic framework for artificial intelligence in marketing, J. Acad. Mark. Sci. 49 (2020) 30–50. doi:10.1007/s11747-020-00749-9.

[23] M. Simon, S. Kamin, A. Hamper, T. Wittenberg, S. Schmitt-Ruth, Strategizing ai in healthcare: A multidimensional blueprint for transformative decision-making in clinical settings, Tech. rep., M. Simon and S. Kamin and A. Hamper and T. Wittenberg and S. Schmitt-Ruth, curr. Dir. Biomed. Eng. (2024). (2024). doi:10.1515/cdbme-2024-2146.

[24] I. Svoboda, D. Lande, Enhancing multi-criteria decision analysis with ai: Integrating analytic hierarchy process and gpt-4 for automated decision support, arXiv preprint arXiv:2402.07404 (2024). doi:10.48550/arXiv.2402.07404.

[25] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, D. Kiela, Retrieval-augmented generation for knowledge-intensive nlp tasks, in: Advances in Neural Information Processing Systems (NeurIPS) 33, 2020.

[26] S. Attari, M. DeKay, C. Davidson, W. B. de Bruin, Public perceptions of energy consumption and savings, Proc. Natl. Acad. Sci. U.S.A. 107 (37) (2010) 16054–16059. doi:10.1073/pnas.1001509107.

[27] C. E. Commission, Ja4.3 walls, Tech. rep., California Energy Commission, 2013 Residential Compliance Manual: Joint Appendix 4 (2013).

[28] U.S. Environmental Protection Agency, U.S. Department of Energy, Recommended home insulation r-values, Tech. rep., ENERGY STAR, based on 2021 IECC Residential Provisions (2023).

[29] U. E. P. Agency, Watersense specification for showerheads, version 1.1, Tech. rep., U.S. Environmental Protection Agency, ePA WaterSense, July 26, 2018. (2018).

[30] U.S. Environmental Protection Agency, Energy star dishwashers: Key product criteria (2023).

[31] U.S. Environmental Protection Agency, Energy star certified residential clothes washers dataset (2023).

[32] U.S. Environmental Protection Agency, Energy star scoping report: Residential clothes dryers, Tech. rep., ENERGY STAR (2023).

[33] J. Dyer, R. Sarin, Measurable multiattribute value functions, Oper. Res. 27 (4) (1979) 810–822. doi:10.1287/opre.27.4.810.

[34] R. Keeney, H. Raiffa, Decisions with Multiple Objectives: Preferences and Value Trade-offs, Wiley, New York, 1976.

[35] PJM Interconnection, 2022 co2 emissions and generation resource mix for pjm and miso, accessed 2025-05-28 (2023).

[36] P. E. Company, How time-of-use pricing works with interconnection to solar or net metering, Tech. rep., PECO Energy Company, tOU customer documentation, 2021. (2021).

[37] E. R. C. of Texas, Impact of increased wind resources in the ercot region, Tech. rep., ERCOT Market Operations and Market Design Departments (2019).

[38] S. P. Holland, E. T. Mansur, N. Z. Muller, A. J. Yates, Spatial and temporal heterogeneity of marginal emissions: Implications for electric cars and other electricity-shifting policies, Tech. rep., National Bureau of Economic Research, nBER Working Paper No. 18462 (2016).

[39] P. Stern, T. Dietz, T. Abel, G. Guagnano, L. Kalof, A value-belief-norm theory of support for social movements: The case of environmentalism, Hum. Ecol. Rev. 6 (2) (1999) 81–97.

[40] F. Fornara, P. Pattitoni, M. Mura, E. Strazzera, Predicting intention to improve household energy efficiency: The role of value-belief-norm theory, normative and informational influence, and specific attitude, J. Environ. Psychol. 45 (2016) 1–10. doi:10.1016/j.jenvp.2015.11.001.

[41] Y. Hong, A. A. Mamun, M. Masukujjaman, Q. Yang, Significance of the environmental value-belief-norm model and its relationship to green consumption among chinese youth, Asia Pac. Manag. Rev. 29 (1) (2024) 127–140. doi:10.1016/j.apmrv.2023.10.002.

[42] L. Steg, L. Dreijerink, W. Abrahamse, Factors influencing the acceptability of energy policies: A test of vbn theory, J. Environ. Psychol. 25 (4) (2005) 415–425. doi:10.1016/j.jenvp.2005.08.003.

[43] S. Lee, Y. Kim, T. Roh, Pro-environmental behavior on electric vehicle use intention: Integrating value-belief-norm theory and theory of planned behavior, J. Clean. Prod. 418 (2023) 138211. doi:10.1016/j.jclepro.2023.138211.

[44] K. Rezaei-Moghaddam, N. Vatankhah, A. Ajili, Adoption of pro-environmental behaviors among farmers: Application of value-belief-norm theory, Chem. Biol. Technol. Agric. 7 (1) (2020). doi:10.1186/s40538-019-0174-z.

[45] C.-P. Wang, Q. zhang, P.P.W. Wong, L. Wang, Consumers' green purchase intention to visit green hotels: A value-belief-norm theory perspective, Front. Psychol. 14 (2023). doi:10.3389/fpsyg.2023.1139116.

[46] H. Lind, T. Nordfjaern, S. Jorgensen, T. Rundmo, The value-belief-norm theory, personal norms and sustainable travel mode choice in urban areas, J. Environ. Psychol. 44 (2015) 119–125. doi:10.1016/j.jenvp.2015.06.001.

[47] E. Roszkowska, Entropy and normalization in mcda: A data-driven perspective on ranking stability, Entropy 28 (1) (2026) 114, pMID: 41594021; PMCID: PMC12839561. doi:10.3390/e28010114.

[48] R. Newell, J. Siikamaki, Nudging energy efficiency behavior: The role of information labels, J. Assoc. Environ. Resour. Econ. 1 (4) (2014) 555–598. doi:10.1086/679281.

[49] H. Allcott, D. Taubinsky, Evaluating behaviorally motivated policy: Experimental evidence from the lightbulb market, Am. Econ. Rev. 105 (8) (2015) 2501–2538. doi:10.1257/aer.20131564.

[50] M. Andor, A. Gerster, S. Sommer, Consumer inattention, heuristic thinking and the role of energy labels, Energy J. 41 (1) (2020) 83–112. doi:10.5547/01956574.41.1.mand.

[51] Field experimental evidence on the effect of pricing on residential electricity consumption, management Science, online ahead of print (2023). doi:10.1287/mnsc.2020.02074.

[52] W. DeOreo, P. Mayer, B. Dziegielewski, J. Kiefer, Residential end uses of water, version 2: Executive report, Tech. rep., Water Research Foundation, project 4309. (2016).

[53] T. H. Poll, Examining americans' shower habits, Tech. rep., The Harris Poll, april 30, 2024. (2024).

[54] N. Aqilah, H. Rijal, S. Zaki, A review of thermal comfort in residential buildings: Comfort trends and energy saving potential, Energies 15 (23) (2022) 9012. doi:10.3390/en15239012.

[55] S. Karjalainen, Gender differences in thermal comfort and use of thermostats in everyday thermal environments, Build. Environ. 42 (4) (2007) 1594–1603. doi:10.1016/j.buildenv.2006.01.009.

[56] Z. Wang, T. Hong, Learning occupants' indoor comfort temperature through a bayesian inference approach for office buildings in united states, Renew. Sustain. Energy Rev. 119 (2020) 109593. doi:10.1016/j.rser.2019.109593.

[57] C. Chen, X. Xu, J. Day, Thermal comfort or money saving? exploring intentions to conserve energy among low-income households in the united states, Energy Res. Soc. Sci. 26 (2017) 61–71. doi:10.1016/j.erss.2017.01.009.

[58] ASHRAE, Standard 55 – thermal environmental conditions for human occupancy, aSHRAE Standard 55.

[59] R. de Dear, G. Brager, Thermal comfort in naturally ventilated buildings: Revisions to ashrae standard 55, Energy Build. 34 (2002) 549–561. doi:10.1016/S0378-7788(02)00005-1.

[60] A. Paetz, E. Dutschke, W. Fichtner, Shifting dish-washer use: A field study on time-of-use tariffs and user acceptance, in: ACEEE Summer Study on Energy Efficiency in Buildings, Paper 0193-000232, 2012, 2012.

[61] M. Waseem, Z. Lin, S. Liu, I. Sajjad, T. Aziz, Optimal gwcso-based home appliances scheduling for demand response considering end-users comfort, Electr. Power Syst. Res. 187 (2020) 106477. doi:10.1016/j.epsr.2020.106477.

[62] C. for Disease Control, Prevention, Monitoring building water: A vital step for control of legionella, Tech. rep., Centers for Disease Control and Prevention, 2026. (2026).

[63] B. Gardner, P. Lally, A. Rebar, Does habit weaken the relationship between intention and behaviour? revisiting the habit-intention interaction hypothesis, Soc. Pers. Psychol. Compass (2020). doi:10.1111/spc3.12553.

[64] E. Mergelsberg, B. Mullan, V. Allom, A. Scott, An intervention designed to investigate habit formation in a novel health behaviour, Psychol. Health 36 (2020) 405–426. doi:10.1080/08870446.2020.1779272.

[65] U. E. I. Administration, Average price of electricity to ultimate customers by end-use sector, by state, 2024 and 2023 (table 2.10), Tech. rep., Electric Power Annual 2024 (2025).

[66] S. E. Economics, A study of heat pump performance in u.s. cities, Tech. rep., Synapse Energy Economics (2013).

[67] D. S. Parker, J. R. Sherwin, T. Stedman, M. Anello, Evaluation of air conditioning performance degradation, Tech. rep., Florida Solar Energy Center, fSEC-CR-2080-18 (2018).

[68] U.S. Environmental Protection Agency, Energy star products, accessed 2025-05-28 (2024).

[69] J. Chen-Yu, J. Emmel, Comparisons of fabric care performances between conventional and high-efficiency washers and dryers, Fash. Text. 5 (2018) 1–19. doi:10.1186/s40691-018-0135-3.

[70] G. Hustvedt, M. Ahn, J. Emmel, The adoption of sustainable laundry technologies by us consumers, Int. J. Consum. Stud. 37 (2013) 291–298. doi:10.1111/ijcs.12007.

[71] D. Winfield, et al., Measured performance of heat pump clothes dryers, in: ACEEE Summer Study on Energy Efficiency in Buildings, 2016, 2016.

[72] N. E. E. Partnerships, New study unearths energy baseline for clothes dryers in the northeast, 2015 (2015).

[73] G. Porras, G. Keoleian, G. Lewis, N. Seeba, A guide to household manual and machine dishwashing through a life cycle perspective, Environ. Res. Commun. 2 (2020). doi:10.1088/2515-7620/ab716b.

[74] R. Hendron, J. Burch, Development of standardized domestic hot water event schedules for residential buildings, Tech. rep., R. Hendron and J. Burch, nREL/TP-550-40874, National Renewable Energy Laboratory, 2008. (2008). doi:10.1115/ES2007-36104.

[75] R. Thaler, Mental accounting matters, J. Behav. Decis. Making 12 (3) (1999) 183–206. doi:10.1002/(SICI)1099-0771(199909)12:3<183::AID-BDM318>3.0.CO;2-F.

[76] C. Heath, J. Soll, Mental budgeting and consumer decisions, J. Consum. Res. 23 (1) (1996) 40–52. doi:10.1086/209465.

[77] D. Prelec, G. Loewenstein, The red and the black: Mental accounting of savings and debt, Mark. Sci. 17 (1) (1998) 4–28. doi:10.1287/mksc.17.1.4.

[78] G. Heutel, Prospect theory and energy efficiency, Tech. rep., G. Heutel, nBER Working Paper No. 23692, National Bureau of Economic Research, 2017. (2017). doi:10.3386/w23692.

[79] J. Gathergood, Self-control, financial literacy and consumer over-indebtedness, J. Econ. Psychol. 33 (3) (2012) 590–602. doi:10.1016/j.joep.2011.11.006.

[80] B. Polly, M. Gestwick, M. Bianchi, Strategy guideline: Accurate heating and cooling load calculations, Tech. rep., National Renewable Energy Laboratory, nREL/TP-5500-51603 (2011).

[81] Air-Conditioning, Heating, and Refrigeration Institute (AHRI), Ansi/ahri standard 210/240-2008: Performance rating of unitary air-conditioning and air-source heat pump equipment (2008).

[82] S. Houde, E. Myers, A. Siddiqui, Smart thermostats, automation, and time-varying prices, Tech. rep., Resources for the Future, rFF Working Paper 21-20 (2021).

[83] ASHRAE, Ansi/ashrae standard 55-2020: Thermal environmental conditions for human occupancy, Tech. rep., ASHRAE (2020).

[84] J. van Hoof, Forty years of fanger's model of thermal comfort: Comfort for all?, Indoor Air 18 (3) (2008) 182–201. doi:10.1111/j.1600-0668.2007.00516.x.

[85] P. Fanger, Thermal Comfort: Analysis and Applications in Environmental Engineering, Danish Technical Press, Copenhagen, 1970.

[86] P. Domanski, H. Henderson, W. Payne, Sensitivity analysis of installation faults on heat pump performance, Tech. rep., P.A. Domanski and H.I. Henderson and W.V. Payne, nIST Technical Note 1848, National Institute of Standards and Technology, 2014. (2014).

[87] D. E. Group, Hvac energy efficiency maintenance study, Tech. rep., Davis Energy Group, prepared for California Investor-Owned Utilities (CALMAC ID HVAC_EE_MAINTENANCE_FINAL), 2010. (2010).

[88] K. Fenaughty, D. Parker, Evaluation of air conditioning performance degradation: Opportunities from diagnostic methods, FSEC-PF-474-18, Florida Solar Energy Center, 2018 (2018).

[89] Air Conditioning Contractors of America (ACCA), Manual j – residential load calculation, Tech. rep., ACCA (2016).

[90] ASHRAE, Ashrae handbook–fundamentals, 2017 edition (2017).

[91] World Health Organization, Environmental noise guidelines for the european region (2018).

[92] U.S. Environmental Protection Agency, Clothes washers – energy star, average US family washes 300 loads/year (25/month). DOE test procedure uses 392/yr (10 CFR 430 App. J1). Accessed 2026-06-10. (2026).

[93] U.S. Department of Energy, Test procedures for clothes dryers and room air conditioners, Tech. rep., U.S. Department of Energy, federal Register 76(4), 2011-01-06 (2010-32118). Revised dryer cycles 416 to 283/year (24/month) per 2005 EIA RECS. (2011).

[94] C. Backman, M. Hoeschele, Validation of a hot water distribution model using laboratory and field data, Tech. rep., National Renewable Energy Laboratory, dOE/GO-102013-39. (2013). doi:10.2172/1220208.

[95] D. Zhang, K.-w. mui, L.-T. Wong, Establishing the relationship between occupants' thermal behavior and energy consumption during showering, Buildings 13 (5) (2023) 1300. doi:10.3390/buildings13051300.

[96] R. M. Company, R. tank water heaters product specifications (electric models 40-55 gal), 2025, Tech. rep., Rheem Manufacturing Company and Residential tank water heaters – product specifications (electric models 40–55 gal) (2025).

[97] M. Shanahan, K. McDonell, L. Reynolds, Role play with large language models, Nature 623 (7987) (2023) 493–498. doi:10.1038/s41586-023-06647-8.

[98] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems (NeurIPS) 35, 2022.

[99] V. Galaz, et al., Machine intelligence, systemic risks, and sustainability, beijer Discussion Paper 274 (2021).

[100] T. Khot, H. Trivedi, M. Finlayson, Y. Fu, K. Richardson, P. Clark, A. Sabharwal, Decomposed prompting: A modular approach for solving complex tasks, in: Proceedings of the 11th International Conference on Learning Representations (ICLR), 2023.

Symbol	Definition
τ	Kendall's rank correlation coefficient
ρ	Spearman's rank correlation coefficient
w_i	weight of criterion i ($i = 1 \ldots 4$), summing to 1
$v_i(x)$	value function score for criterion i at raw value x, range $[0, 10]$
α	shape parameter for logarithmic value functions
s_j	MAVT weighted sum score for alternative j
MAE	Mean Absolute Error on MAVT score (0–10 scale)
RMSE	Root Mean Squared Error on MAVT score (0–10 scale)
n	number of test scenarios evaluated
k	number of retrieved chunks per query in RAG ($k = 3$)

Table 7: Notation used throughout this paper.

Criterion	Weight (%)	Value Function	α	Direction
Environmental Impact	35	Linear	—	Lower raw = better
Energy Cost	30	Linear	—	Lower raw = better
Comfort	20	Logarithmic	1.5	Higher raw = better
Practicality	15	Logarithmic	1.2	Higher raw = better

Table 8: MAVT criterion specification. Weights sum to 100% and are held constant across all architectures and the ground truth calculator.

$$v_i(x) = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \times 10 \tag{25}$$

$$v_i(x) = \frac{\ln\left(1 + \alpha \cdot \frac{x - x_{\min}}{x_{\max} - x_{\min}}\right)}{\ln(1 + \alpha)} \times 10 \tag{26}$$

Decision Type	Test Set (n)	RAG Set (n)	Total (n)
HVAC	70	35	105
Appliance	65	35	100
Shower	60	20	80
Total	195	90	285

Decision Type	GT Calculator Receives	Withheld / Generalized for AI
HVAC	R-value, SEER, HVAC age, occupancy context, sq. ft., insulation	R-value, SEER, HVAC age, occupancy context (insulation shown as good/avg/poor)
Appliance	kWh/cycle, TOU rates, baseline time, appliance type	kWh/cycle, TOU rates (type and baseline time inferred from question text)
Shower	GPM, tank size, water heater setpoint	GPM (from flow-rate label), tank size, water heater setpoint

Table 9: Left: scenario distribution (BLOCKED in this draft; numbers stale – see audit log). Right: parameters available to the ground-truth calculator but withheld or generalized for AI input.

Metric	Pure Prompting	RAG-Enhanced	Hybrid
Ranking Accuracy			
Kendall's τ	-0.06	0.43	**0.80**
Spearman's ρ	-0.055	0.49	**0.82**
Top-1 Accuracy (%)	30.00%	58.00%	**83.00%**
Top-2 Accuracy (%)	62.00%	89.00%	**97.00%**
Score Error			
Overall MAE	2.52	1.61	**0.69**
Overall RMSE	3.07	2.40	**1.52**

Numbers BLOCKED in this draft; see audit log.

Table 10: Overall performance comparison (BLOCKED – stale numbers).

Criterion	Pure Prompting	RAG-Enhanced	Hybrid
Energy Cost	2.60	1.60	**0.96**
Environmental Impact	2.80	1.59	**1.02**
Comfort	2.51	1.97	**0.32**
Practicality	2.16	1.29	**0.43**
Overall	2.52	1.61	**0.69**

Numbers BLOCKED in this draft. Environmental MAE for Shower reflects water-consumption proxy (gallons); see Section 3.3.

Table 11: MAE per criterion by architecture (BLOCKED – stale numbers).

	Pure Prompting			RAG-Enhanced			Hybrid		
Decision Type	τ	Top-1 (%)	Top-2 (%)	τ	Top-1 (%)	Top-2 (%)	τ	Top-1 (%)	Top-2 (%)
HVAC	-0.11	25.00%	56.67%	0.52	63.33%	93.33%	**0.90**	**91.67%**	100.00%
Appliance	0.09	40.00%	72.00%	0.41	56.00%	88.00%	**0.68**	**72.00%**	92.00%
Shower	-0.11	33.33%	66.67%	0.11	40.00%	73.33%	**0.60**	**66.67%**	93.33%
Overall	-0.06	30.00%	62.00%	0.43	58.00%	89.00%	**0.80**	**83.00%**	97.00%

Table 12: Ranking accuracy by decision type (BLOCKED – stale numbers).

Metric	Pure Prompting	RAG-Enhanced	Hybrid
Total API Calls	300	300	100
Input Tokens	114739	295174	81741
Output Tokens	9970	13463	16275
Total Tokens	124709	308637	**98016**
Total Latency (s)	437.1	480.8	**275.6**
Avg. Latency/Call (ms)	1456.9	1602.6	2755.9
Success Rate (%)	100.00%	100.00%	100.00%
Failure Rate (%)	0.00%	0.00%	0.00%

Table 13: API usage and efficiency (BLOCKED – stale numbers).

Side-by-side vertical pipeline flowcharts for three LLM integration architectures. Pure Prompting: 3 LLM calls, one per alternative, no external context. RAG-Enhanced: 3 LLM calls each preceded by ChromaDB retrieval ($k = 3$ chunks). Hybrid: 1 LLM call for parameter extraction, output to physics-based calculator, then MAVT weighted sum.

Figure 4: Processing pipeline for each architecture. Pure Prompting and RAG-Enhanced require three API calls per scenario; Hybrid requires one.